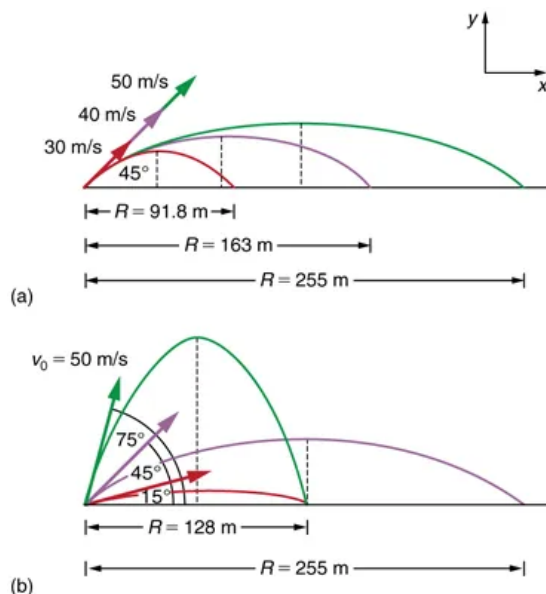


Chapter 3 Homework B (Projectiles)



$$\vec{v} = \vec{v}_0 + \vec{a}t$$

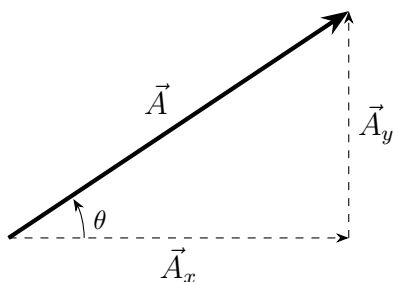
"Old Faithful"

$$\vec{x} = \vec{x}_0 + \vec{v}_0t + \frac{1}{2}\vec{a}t^2$$

"The Big Chalupa"

$$\vec{v}^2 = \vec{v}_0^2 + 2\vec{a} \cdot (\vec{x} - \vec{x}_0)$$

"Ain't Got No Time"



$$A_x = A \cos \theta$$

$$A_y = A \sin \theta$$

$$\theta = \tan^{-1} \left(\frac{A_y}{A_x} \right)$$

$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

$$\vec{v}_{AC} = \vec{v}_{AB} + \vec{v}_{BC}$$

$$\vec{v}_{AB} = -\vec{v}_{BA}$$

Selected Answers

- 12.5 s; 50 m
- (a) 7 busses; (b) margin of error is about 6.7 m
- (a) 560 m/s; (b) 8000 m
- (a) 18.4°; (b) max height is 6.23 m, so it makes it over the branch
- 22.3 m
- 6.65 m/s
- 2.70 m/s east; 0.30 m/s east
- (a) 10.5 m/s; (b) 6.5 m/s
- 1.66 m/s @ 65.0° E of N
- 8.05 m/s @ 81.8° N of E
- 7.7 m/s
- 10.3 m/s @ 73.1° below horiz
- No. Max range $\approx$ 92 m
- 2.16 m/s at 33.7° east of north
- 228 km/h at 15.3° west of north
- 25 s
- 1.50 m
- 8.05 m/s @ 81.8° N of E

Name:

Date:

Period:

Questions and figures are from *OpenStax College Physics 2nd ed (Chapter 3)* or *Giancoli Physics: Principles with Applications 7th ed. (Chapter 3)* unless otherwise noted.

Projectile Motion (Part II)

Your work for these problems should include diagrams and knowns/unknowns.

- (Giancoli P26) Extreme-sports enthusiasts have been known to jump off the top of El Capitan, a sheer granite cliff of height 910 m in Yosemite National Park. Assume a jumper runs horizontally off the top of El Capitan with speed 4.0 m/s and enjoys a free fall until she is 150 m above the valley floor, at which time she opens her parachute (Figure 1). (a) How long is the jumper in free fall? Ignore air resistance. (b) It is important to be as far away from the cliff as possible before opening the parachute. How far from the cliff is this jumper when she opens her chute?
- (OpenStax P28) (a) A daredevil is attempting to jump his motorcycle over a line of buses parked end to end by driving up a 32° ramp at a speed of 40.0 m/s. How many buses can he clear if the top of the takeoff ramp is at the same height as the bus tops and the buses are 20.0 m long? (b) Discuss what your answer implies about the margin of error in this act—that is, consider how much greater the range is than the horizontal distance he must travel to miss the end of the last bus. (Neglect air resistance.)
- (OpenStax P33) The cannon on a battleship can fire a shell a maximum distance of 32.0 km. (a) Calculate the initial velocity of the shell. (b) What maximum height does it reach? (At its highest, the shell is above 60% of the atmosphere—but air resistance is not really negligible as assumed to make this problem easier.)

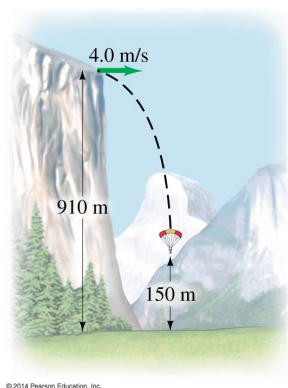


Figure 1: Fig 3-37 from Giancoli

4. (OpenStax P29) An archer shoots an arrow at a 75.0 m distant target; the bull's-eye of the target is at same height as the release height of the arrow. (a) At what angle must the arrow be released to hit the bull's-eye if its initial speed is 35.0 m/s? (b) There is a large tree halfway between the archer and the target with an overhanging horizontal branch 3.50 m above the release height of the arrow. Will the arrow go over or under the branch?
5. (Giancoli P29) A shot-putter throws the "shot" (mass 73 kg) with an initial speed of 14.4 m/s at a 34.0° angle to the horizontal. Calculate the horizontal distance traveled by the shot if it leaves the athlete's hand at a height of 2.10 m above the ground.
6. (Giancoli P56) Romeo is throwing pebbles gently up at Juliet's window, and he wants the pebbles to hit the window with only a horizontal component of velocity. He is standing at the

edge of a rose garden 8.0 m below her window and 8.5 m from the base of the wall (Figure 2). How fast are the pebbles going when they hit her window?

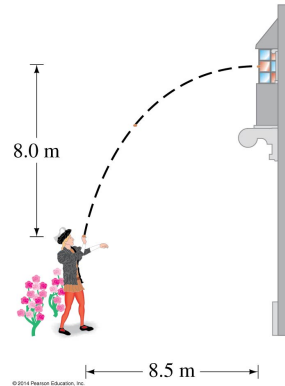


Figure 2: Fig 3-49 from Giancoli

[See #17-#19 for additional practice]

Please answer each question in at least one complete sentence

- (Original question) A projectile is launched at an angle 25° above horizontal. When (if ever) is its velocity zero?
- (OpenStax CQ15) For a fixed initial speed, the range of a projectile is determined by the angle at which it is fired. For all but the maximum, there are two angles that give the same range. Considering factors that affect the ability of an archer to hit a target, such as wind, explain why the smaller angle (closer to the horizontal) is preferable. On the other hand, why does the punter in a football game use the higher trajectory?
- (Giancoli CQ15) A projectile is launched at an upward angle of 30° to the horizontal with a speed of 30 m/s. How does the horizontal component of its velocity 1.0 s after launch compare with its horizontal component of velocity 2.0 s after launch, ignoring air resistance? Explain.

Relative Motion

10. (Original Problem) A moving walkway travels east at 1.50 m/s . A student walks east on it at 1.20 m/s relative to the walkway, then turns and walks west at the same speed relative to the walkway. Find the student's velocity relative to the ground in each case.
11. (Giancoli P38) A jogger is exercising on the deck of a cruise ship. The ship moves forward at a constant speed of 8.5 m/s relative to the water.
- While running toward the bow (front) of the ship, the jogger's speed relative to the ship is 2.0 m/s . Determine the jogger's velocity relative to the water.
 - Later, the jogger runs toward the stern (rear) of the ship at the same speed relative to the ship. What is the jogger's velocity relative to the water in this case?
12. (Giancoli P39) Huck Finn stands on a raft drifting down the Mississippi River. The raft moves at 1.50 m/s relative to the riverbank. Huck walks at 0.70 m/s relative to the raft, moving perpendicular to the raft's motion (that is, across the raft). Determine Huck's velocity

(both magnitude and direction) relative to the riverbank.

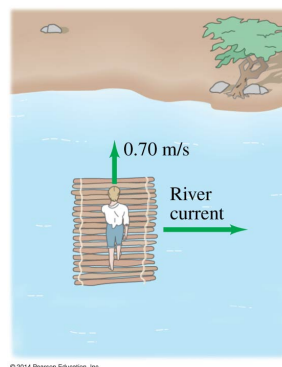


Figure 3: Fig 3-42 from Giancoli. Relative velocity of Huck walking across a drifting raft.

13. (OpenStax P57) A ship sets sail from Rotterdam, The Netherlands, heading due north at 7.00 m/s relative to the water. The local ocean current is 1.50 m/s in a direction 40.0° north of east. What is the velocity of the ship relative to the Earth?

[See #20-#22 for additional practice]

Relative Velocity Conceptual Questions

14. (OpenStax CQ19) If someone is riding in the back of a pickup truck and throws a softball straight backward, is it possible for the ball to fall straight down as viewed by a person standing at the side of the road? Under what condition would this occur? How would the motion of the ball appear to the person who threw it?
15. (OpenStax CQ20) The hat of a jogger running at constant velocity falls off the back of his head. Draw a sketch showing the path of the hat in the jogger's frame of reference. Draw its path as viewed by a stationary observer.
16. (Giancoli CQ19) If you are riding on a train that speeds past another train moving in the same direction on an adjacent track, it appears that the other train is moving backward. Why?

Additional Practice

These problems are provided for extra practice. They are *not required* and there is no bonus for doing them.

17. (Giancoli P20) A ball is thrown horizontally from the roof of a building 7.5 m tall and lands 9.5 m from the base. What was the ball's initial speed?
18. (OpenStax P40) An eagle is flying horizontally at a speed of 3.00 m/s when the fish in her talons wiggles loose and falls into the lake 5.00 m below. Calculate the velocity of the fish relative to the water when it hits the water.
19. (OpenStax P43) Can a goalkeeper at her/his goal kick a soccer ball into the opponent's goal without the ball touching the ground? The distance will be about 95 m. A goalkeeper can give the ball a speed of 30 m/s.
20. (Original Problem) A swimmer can swim 1.80 m/s north relative to the water. A river current flows 1.20 m/s east. Find the swimmer's velocity relative to the shore.
21. (Original Problem) A plane has an air speed of 220 km/h due north. The wind blows 60.0 km/h due west. Find the plane's ground speed and direction.
22. (Giancoli P41, modified) Two planes approach each other head on. Each has a speed of 200 m/s, and they spot each other when they are initially 10 km apart. How much time to the airplanes have to take evasive action?

Bonus

You will be given bonus if you complete these problems *on a separate sheet of paper*

23. (OpenStax P35) In the standing broad jump, one squats and then pushes off with the legs to see how far one can jump. Suppose the extension of the legs from the crouch position is 0.600 m and the acceleration achieved from this position is 1.25 times the acceleration due to gravity, g . How far can they jump? State your assumptions. (Increased range can be achieved by swinging the arms in the direction of the jump.)
24. (OpenStax P47) A football player punts the ball at a 45.0° angle. Without an effect from the wind, the ball would travel 60.0 m horizontally. (a) What is the initial speed of the ball? (b) When the ball is near its maximum height it experiences a brief gust of wind that reduces its horizontal velocity by 1.50 m/s. What distance does the ball travel horizontally?
25. (Giancoli P42) A boat moves across a still lake at 1.70 m/s. A passenger walks up a flight of stairs on the boat at a speed of 0.60 m/s relative to the boat. The stairs are oriented at a 45°

angle in the forward direction, as shown in Figure 4. Determine the velocity of the passenger relative to the water.

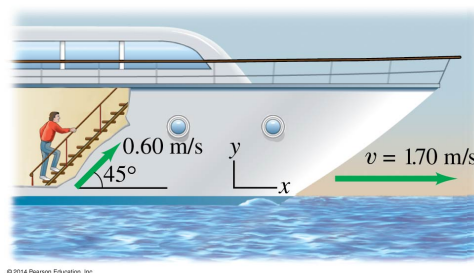


Figure 4: Fig 3-43 from Giancoli

26. (Giancoli P50) An airplane has an airspeed of 580 km/h and must fly along a straight path directed 38° north of east. A steady wind of 82 km/h blows from the north. In what direction should the airplane head in order to maintain the required path?